1. Basic Prompt writing: Create two different prompts to ask an AI about the topic "Electricity." The first prompt should be vague, and the second prompt should be clear and specific. Compare the responses you get and describe which prompt gave a better answer and why.

Prompt 1: Vague Prompt

Q: Explain electricity.

Response:
Electricity is a form of energy produced by the flow of electric charges (electrons) that is used to power devices and systems in everyday life.

Prompt 2: Clear and Specific Prompt

Q: Explain electricity in simple terms, including its definition, methods of generation, and three practical applications in daily life.

Electricity is a type of energy created by the flow of electric charges, mainly electrons, through wires or conductors. It is one of the most important energy sources used in daily life for homes, schools, industries, and transportation. Electricity is generated from different sources such as water in hydroelectric plants, heat in thermal power stations, and sunlight using solar panels. It can also be produced from wind energy. In daily life, electricity is used for lighting houses and streets, operating appliances like fans, TVs, and refrigerators, and charging electronic devices such as phones, laptops, and batteries.

Comparison and Analysis:

The second prompt produced a more effective response because it was specific and clearly defined. It instructed the AI to include the meaning, generation methods, and applications of electricity, resulting in a structured and detailed answer. The first prompt was vague and broad, so the response was more general. This shows that specific prompts help generate clearer, more relevant, and more informative responses.

2. **Zero-Shot Prompting**: Create a prompt that asks an AI to explain Ohm's Law without giving any example or background. Evaluate how well the AI explains the concept based on your prompt alone.

Q: Explain Ohm's Law

AI Response:

Ohm's Law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided the temperature and physical conditions remain constant.

It is expressed mathematically as:

$V = I \times R$

Where:

- **V** = Voltage (potential difference)
- **I** = Current (flow of electric charge)
- **R** = Resistance (opposition to current flow)

In simple terms, if voltage increases, current also increases (when resistance stays the same), and if resistance increases, current decreases.

Evaluation of AI Response: Ohm's Law

The AI response to Ohm's Law is accurate, clear, and well-structured. It correctly defines the law, presents the formula $V = I \times R$, and explains the relationship between voltage, current, and resistance in simple terms. The explanation follows a logical flow and meets the prompt requirement effectively. However, it lacks deeper insight, such as real-life applications or conditions where the law may not apply. Despite this, it is suitable for beginners and provides a strong conceptual understanding.

3. One-Shot and Few-Shot Prompting: Provide the AI with a single example of how to calculate the resistance in a simple circuit. Then write your own prompt asking the AI to solve a similar resistance calculation. After that, add two more examples to your prompt and observe any changes in the AI's response quality.

Topic: Resistance Calculation Using Prompting Techniques

Concept used: Ohm's Law

Formula Used:

R = V / I

Where:

- R = Resistance (Ω)
- V = Voltage (V)
- I = Current (A)

One-Shot Prompting:

Example Given:

V = 12V, I = 3A → R = 12 / 3 = 4 Ω

Prompt:

Q: In a circuit with 20V voltage and 5A current, find resistance.

AI Output:

R = 20 / 5 = 4 Ω

Result:

- Correct answer
- Works with single example
- Moderate understanding of pattern

Few-Shot Prompting:

Examples Given:

1. V = 12V, I = 3A → R = 4 Ω
2. V = 20V, I = 5A → R = 4 Ω

Prompt:

Q: In a circuit with 30V voltage and 6A current, find resistance.

AI Output:

R = 30 / 6 = 5 Ω

Result:

- Faster pattern recognition
- More consistent format
- Higher accuracy and confidence

Efficiency Comparison:

Feature	One-Shot	Few-Shot
Learning speed	Medium	High
Pattern understanding	Basic	Strong
Accuracy consistency	Moderate	High
Clarity of output	Good	Excellent

Final Conclusion:

Few-shot prompting is **more efficient and reliable** because multiple examples help the AI quickly learn the pattern of solving resistance problems using Ohm's Law. It reduces ambiguity and improves consistency compared to one-shot prompting.

4. Chain-of-Thought Prompting: Develop a prompt that guides the AI step-by-step through calculating current flow in a circuit using Ohm's Law with resistors in series. Then, ask a final question for the AI to solve. Analyse how breaking down the reasoning steps impacts the accuracy of the answer.

Chain-of-Thought Prompt (Step-by-Step Instruction):

Prompt:

Q: Calculate the current flowing in a series circuit step by step. First, find the total resistance of the circuit, then apply Ohm's Law to find the current. Assume two resistors in series: 4Ω and 6Ω, and a voltage supply of 20V.

Step-by-Step Solution (AI Reasoning Flow):

Step 1: Find Total Resistance (Series Rule)

In a series circuit:
R(total) = R1 + R2

So,
R(total) = 4Ω + 6Ω = **10Ω**

Step 2: Apply Ohm's Law

Formula:
I = V / R

Substitute values:
I = 20V / 10Ω = **2A**

Final Answer:

Current flowing in the circuit = 2A

Analysis of Chain-of-Thought Prompting:

✓Improved Accuracy

Breaking the problem into steps reduces calculation mistakes and ensures correct application of formulas.

✓ Better Logical Flow

The model clearly separates:

- Resistance calculation
- Current calculation

✓ Easier Verification

Each step can be checked individually, making error detection simple.

✓ Strong Concept Understanding

Helps reinforce how Ohm's Law is applied in real circuit problems.

Final Conclusion:

Chain-of-thought prompting significantly improves **accuracy, clarity, and reasoning structure** by forcing step-by-step computation. It reduces confusion in multi-step electrical problems and ensures correct final answers through logical breakdown.

5.Prompt Refinement: Start with an ambiguous prompt related to the "Water Cycle." Test the AI's response, note the confusion or errors, and then refine your prompt to make it clearer and more specific. Repeat this process twice and record how the AI's responses improve with each refinement.

Role-Based Prompting: Create three prompts asking the AI to explain "Newton's Laws of Motion," each with a different role instruction: (a) as an expert engineer, (b) as a high school teacher, (c) as a beginner. Compare the tone, detail, and style of the responses.

Part 1: Prompt Refinement (Water Cycle):

Concept: Water Cycle

Step 1: Ambiguous Prompt

Prompt: Explain the water cycle

Response:
The water cycle is the movement of water on Earth involving evaporation, condensation, and precipitation.

Issues:

- Very brief
- Missing process flow
- No clear structure

Step 2: First Refinement

Prompt:
Explain the water cycle with its main stages

Response:
The water cycle includes evaporation, condensation, and precipitation. Water moves between land, air, and oceans continuously.

Improvement:

- Better structure
- Still lacks detailed flow and connection between stages

Step 3: Second Refinement

Prompt:
Explain the water cycle clearly by describing evaporation, condensation, precipitation, and collection in sequence

Response:
The water cycle is the continuous movement of water. Evaporation changes water into vapor, condensation forms clouds, precipitation brings water back as rain, and collection gathers water in rivers and oceans, repeating the cycle.

Final Improvement:

- Clear step-by-step flow
- Logical sequence
- Complete explanation

Conclusion (Efficiency View)

- More specific prompts improve clarity and completeness
- Structured instructions reduce ambiguity
- Final refined prompt gives the most accurate and well-organized response

Part 2: Role-Based Prompting (Newton's Laws):

Concept: Newton's Laws of Motion

(a) Expert Engineer Role

Prompt: Explain Newton's Laws of Motion as an expert engineer

Response Style:

- Highly technical

- Includes equations and engineering applications
- Focus on real-world mechanical systems

(b) High School Teacher Role

Prompt: Explain Newton's Laws of Motion as a high school teacher

Response Style:

- Simple structured explanation
- Easy language
- Basic examples for students

(c) Beginner Role

Prompt: Explain Newton's Laws of Motion for a beginner

Response Style:

- Very simple and conversational
- Minimal technical terms
- Focus on understanding concepts

Comparison Summary

- Expert role → technical, advanced, application-based
- Teacher role → balanced, clear, educational
- Beginner role → simplest, intuitive explanation

Final Conclusion:

Role-based prompting improves control over AI responses by adjusting **depth, tone, and complexity**. It makes the same concept adaptable for different audiences, improving communication efficiency.

6.**Creative Engineering Problem Prompts**: Craft a prompt that asks the AI to brainstorm ideas for designing a low-cost water purification system suitable for rural areas. Encourage creativity by adding phrases like “limited resources” and “sustainability”.

Creative Engineering Prompt:

Topic: Low-cost water purification system for rural areas

Concept focus: Water Purification

Prompt:

“Act as an innovative engineering designer. Brainstorm practical and creative ideas for designing a **low-cost water purification system** suitable for rural areas with **limited resources and infrastructure**. Ensure the solution is **sustainable, easy to maintain, and environmentally friendly**. Consider locally available materials, energy efficiency, and simple purification methods that do not require advanced technology. Provide multiple design ideas and briefly explain how each one works.”

Expected AI Output (What this prompt encourages):

- Simple, affordable filtration models
- Use of natural materials like sand, charcoal, gravel
- Solar-based or gravity-based systems
- Community-level water treatment ideas
- Focus on sustainability and maintenance ease

Why this prompt is effective:

- Clearly defines constraints (limited resources, rural setting)
- Encourages creativity and multiple solutions
- Guides AI toward practical engineering thinking
- Ensures focus on sustainability and real-world usability

7.Ethical Prompt Design Discussion: Identify a biased prompt related to job descriptions (e.g. language with respect to a gender). Rewrite the prompt to remove bias and create a neutral, inclusive version. Explain why this revision is more ethical**.**

Biased Prompt (Example):

"Describe the qualities of a strong female nurse who is caring, gentle, and emotionally soft."

Issues in the Prompt:

- It assumes certain traits are linked to a specific gender
- It reinforces stereotypes (e.g., women being "emotionally soft")
- It limits professional qualities based on identity rather than skills
- It is not inclusive and can exclude or misrepresent people

Rewritten Neutral & Inclusive Prompt:

"Describe the qualities of a strong nurse, focusing on professional skills, competencies, and personal attributes required for effective patient care."

Why the Revised Prompt is More Ethical:

- Removes any gender reference, making it inclusive for all individuals
- Focuses on **skills and professional competencies** rather than stereotypes
- Avoids reinforcing biased assumptions about personality traits
- Promotes fairness by allowing any qualified person to be represented equally
- Encourages objective and respectful language in AI responses

Conclusion:

The revised prompt is more ethical because it ensures neutrality, inclusivity, and fairness while describing job roles. It prevents biased assumptions and focuses only on relevant professional qualities.

8.**Simulated Customer Support Chatbot**: Develop a prompt that instructs the AI to play the role of a technical support agent helping a customer troubleshoot a failure in an electronic circuit. Include instructions to keep the tone friendly and professional and to ask diagnostic questions.

Prompt:

Act as a technical support agent for an electronics service center. Your role is to help a customer troubleshoot a failure in an electronic circuit. Communicate in a friendly, polite, and professional manner. Ask diagnostic questions step by step to identify the issue before suggesting a solution. Focus on practical troubleshooting, such as checking power supply, loose connections, damaged components, and measurement readings. Provide clear explanations in simple language suitable for an engineering student or beginner. After identifying the likely problem, suggest possible solutions and safety precautions.

Example interaction:

Customer: My circuit is not working after switching on the power supply.

AI Support Agent: Hello, I'll help you troubleshoot the issue. Let's check a few things step by step.

1. Is the power supply output voltage normal when measured with a multimeter?
2. Are all connections properly fixed, or do you notice any loose wires?
3. Do any components such as Resistor, capacitor, or transistor appear burned or overheated?
4. Is the LED indicator or output device turning on at all?
5. Have you recently changed any component values or wiring connections?

Based on your answers, I can help identify whether the issue is due to power supply failure, wrong connection, or component damage.

9. **Multi-Language Prompting:** Develop a prompt that asks the AI to translate a simple engineering glossary (5 technical terms) from English to your native language. Then modify the prompt to request additional explanations of these terms in the translated language.

Basic Translation Prompt:

Prompt:
Translate the following engineering terms from English into Kannada:

1. Voltage
2. Current
3. Resistance
4. Power
5. Circuit

Enhanced Prompt with Explanations:

Prompt:
“Translate the following engineering terms from English into Kannada and also provide a simple explanation for each term in Kannada:

1. Voltage
2. Current
3. Resistance
4. Power
5. Circuit

Ensure the explanations are easy to understand and suitable for beginners.”

AI Response:

English	Kannada	Explanation
Voltage	ವೋಲ್ಟೇಜ್	Electrical pressure
Current	ವಿದ್ಯುತ್ ಪ್ರವಾಹ	Flow of charge
Resistance	ಪ್ರತಿರೋಧ	Opposition to current
Power	ಶಕ್ತಿ	Rate of energy usage
Circuit	ವಲಯ	Closed path for current

10. Review a curated set of different prompt types (e.g., for summarization, information extraction, paraphrasing, question answering) from a “Prompt Gallery.” For each prompt type, match it with a real- world task (e.g., summarizing a lecture note, extracting names from a project report). Test at least three prompt templates on an AI tool or by role-play (students simulate being the AI), with varied wording. Record the outcomes and discuss which prompt (or template) was most effective for each task, and explain why you think it worked best. Reflect on how changing small parts of a prompt can alter model response quality, completeness, or accuracy.

Objective:

To study how different prompt types affect AI responses and understand how wording changes output quality, completeness, and accuracy.

A) Summarization Prompt:

Real-world task:

Summarizing class lecture notes on climate change.

Source text:

Climate change refers to long-term changes in Earth’s temperature and weather patterns. Human activities such as burning fossil fuels increase greenhouse gases, causing global warming. This affects rainfall, sea levels, and ecosystems.

Tested Prompt Templates:-

Template 1: Summarize the paragraph.
Outcome: Climate change is caused by global warming and affects the environment.

Template 2: Summarize the paragraph in 2–3 sentences covering all main points.
Outcome: Climate change is a long-term shift in weather patterns. Human

activities release greenhouse gases that cause global warming and impact sea levels, rainfall, and ecosystems.

Template 3: Provide a concise academic summary of the paragraph.
Outcome: Climate change is a global environmental issue driven mainly by human-induced greenhouse gas emissions, affecting climate systems and biodiversity.

Most effective: Template 2

Reason:

The prompt clearly defined the expected length and requested all key points. This produced a balanced summary without losing important details.

B) Information Extraction Prompt:

Real-world task:

Extracting key details from a project report.

Source text:

The Smart Attendance System was developed by Ananya Rao and Kiran Patel on 10 March 2026 under the supervision of Prof. Ramesh Kumar.

Tested Prompt Templates:-

Template 1: Extract important information from the text.
Outcome: Smart Attendance System, Ananya Rao, Kiran Patel, 10 March 2026, Prof. Ramesh Kumar

Template 2: Identify project title, people involved, and date.
Outcome: Project: Smart Attendance System
People: Ananya Rao, Kiran Patel, Prof. Ramesh Kumar
Date: 10 March 2026

Template 3: Extract the project name, contributors, guide, and completion date in structured format.
Outcome: Project Name: Smart Attendance System
Contributors: Ananya Rao, Kiran Patel

Guide: Prof. Ramesh Kumar
Date: 10 March 2026

Most effective: Template 3

Reason:

The structured request produced categorized output, which was easier to read and more suitable for documentation.

C) Paraphrasing Prompt:

Real-world task:

Rewriting technical notes in simple words.

Source sentence:

Machine learning enables computers to learn from data and improve performance without explicit programming.

Tested Prompt Templates:-

Template 1: Rewrite this sentence.
Outcome: Machine learning helps computers learn from data without direct programming.

Template 2: Paraphrase in simple words.
Outcome: Machine learning allows computers to study data and improve by themselves.

Template 3: Rewrite using different wording while keeping the exact meaning.
Outcome: Machine learning allows systems to analyze data and enhance their performance without being directly programmed.

Most effective: Template 3

Reason:

It preserved the original meaning accurately while changing the sentence structure effectively.

D) Question Answering Prompt:

Real-world task:

Answering a textbook science question.

Question:

What is the function of the human heart?

Tested Prompt Templates:-

Template 1: What is the function of the human heart?
Outcome: The heart pumps blood through the body.

Template 2: Explain the function of the human heart in 2 sentences.
Outcome: The human heart pumps blood to all parts of the body. It supplies oxygen and nutrients through circulation.

Template 3: Answer as if explaining to a school student.
Outcome: The heart is an organ that pumps blood around the body, carrying oxygen and nutrients to keep us alive and healthy.

Most effective: Template 3

Reason:

Mentioning the target audience improved clarity and made the explanation more understandable.

Comparative Observation:

Prompt Type	Best Template	Key Benefit
Summarization	Template 2	Better completeness
Information Extraction	Template 3	Organized output
Paraphrasing	Template 3	Preserved meaning
Question Answering	Template 3	Audience-focused explanation

Reflection:

The experiment showed that prompt design directly influences AI performance. Small changes such as specifying output format, response length, or target audience improved response quality significantly.

Prompts with detailed instructions produced more accurate and useful answers, while short generic prompts often resulted in incomplete responses. Structured prompts guided the AI more effectively and reduced ambiguity.

This activity demonstrates that prompt engineering is important because even minor wording adjustments can improve AI-generated results in practical tasks such as summarization, extraction, rewriting, and answering questions.

11. Choose a real engineering challenge or societal problem relevant to your field (e.g., “Reducing plastic waste in campus cafeterias” or “Optimizing solar panel placement on campus rooftops”). Draft an initial prompt that asks an AI to propose practical solutions. Share the AI’s (or peer’s) answer in small groups and identify aspects that are missing, vague, or not actionable. Refine your prompt based on feedback (e.g., specify constraints, ask for step-by-step solutions, or require a list of pros and cons). Repeat the process one more time, refining again for further clarity or specificity. Document the entire prompt-refinement process and share the best solution generated, along with a brief analysis of how prompt improvements led to better responses.

Chosen Problem:

Improving classroom energy efficiency by reducing unnecessary electricity usage in college smart classrooms.

Round 1: Initial Prompt

Initial Prompt:
Suggest solutions to reduce electricity waste in smart classrooms in colleges.

AI Response:

- Turn off lights when not needed
- Use LED bulbs
- Use solar power
- Install smart switches

Issues Found:

- Response was too general
- No implementation steps

- No cost or feasibility mentioned
- No advantages/disadvantages
- Not specific to college environment

Round 2: Refined Prompt

Refined Prompt:
Suggest practical solutions to reduce electricity waste in college smart classrooms. Include low-cost methods, automation ideas, and implementation steps.

AI Response:

1. Install motion sensors for lights and fans
2. Replace normal lights with LED systems
3. Use smart meters to monitor power usage
4. Schedule automatic shutdown after class hours

Issues Found:

- Better than first response
- Still lacked detailed explanation
- No comparison between methods
- No clear recommendation for best option

Round 3: Final Refined Prompt

Final Prompt:
Suggest practical and low-cost solutions to reduce electricity waste in college smart classrooms. For each solution, provide implementation steps, required components, advantages, disadvantages, and the most suitable method for a medium-sized engineering college campus.

Final AI Response:

Solution 1: Motion Sensor-Based Control

Components:

- PIR motion sensor
- Relay module
- Microcontroller (Arduino Uno)
- Smart switch

Steps:

1. Install PIR sensors in classroom
2. Connect to controller
3. Detect room occupancy
4. Automatically control lights/fans
5. Turn off devices when room is empty

Benefits:

- Saves electricity automatically
- Low maintenance
- Efficient for classrooms

Limitations:

- Initial setup cost
- Sensor calibration required

Solution 2: Smart Energy Monitoring

Components:

- Smart meter
- Wi-Fi module
- Mobile dashboard

Steps:

1. Install digital meter

2. Connect monitoring system
3. Track daily usage
4. Generate reports
5. Analyse wastage patterns

Benefits:

- Real-time monitoring
- Data analysis possible

Limitation:

- Requires internet
- Slightly higher cost

Solution 3: Solar-Assisted Backup System

Components:

- Solar panel
- Battery
- Inverter

Steps:

1. Install rooftop solar panels
2. Connect battery backup
3. Supply power to classrooms

Benefits:

- Renewable source
- Long-term savings

Limitations:

- High installation cost
- Maintenance needed

Best Solution

Motion Sensor-Based Control System

This solution was most suitable because it is cost-effective, easy to implement, and directly reduces electricity wastage without requiring major infrastructure changes.

Analysis of Prompt Improvement:

The response quality improved significantly as the prompt became more specific

Improvement Observed:

Prompt Version	Quality	Observed Change
Initial	Low	Basic ideas only
Improved	Medium	Added technical suggestions
Final	High	Detailed, structured engineering solution

Key Learning:

Small changes in prompt wording greatly improved the usefulness of AI responses.

Adding details such as:

- target environment
- constraints
- required output format
- evaluation criteria
- recommendation request

made the response more practical and engineering-oriented.

The final prompt produced the best result because it clearly defined expectations and guided the AI toward a complete problem-solving approach.